

THz electronics enabled by cost-effective germanium semiconductors

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Abstract—The operation frequencies of modern electronics are steadily increasing towards the THz frequency range. THz devices have been traditionally based on materials that are grown epitaxially, methods that are expensive and not widely available at scale. Here, we present a cost-effective alternative for the development of THz electronic and opto-electronic devices based on evaporated amorphous germanium. Our results show that thin layers of Ge, deposited at moderate temperatures, exhibit sub-picosecond dynamics and can be used to generate picosecond pulses through photo-conductive switches.

I. INTRODUCTION

HIGH-speed electronics are projected to become the backbone of future technologies, such as ultrafast telecommunications, quantum technologies and fast power electronics, and are on track to transform healthcare and security applications, while providing new fundamental research directions. Over the past years, operation frequencies of electronics have been steadily increasing from the ‘conventional’ GHz range towards the underutilised Terahertz (THz) frequency domain. With such high frequencies, even though they can offer solutions to current technological problems, it is still challenging to effectively deploy them at scale, due to the lack of suitable and cost-effective sources and detectors. Charge carrier dynamics are fundamental for semiconductor-based THz technologies. Controlling recombination times and carrier mobilities at picosecond (ps) timescales is essential for applications requiring rapid signal processing, including ultrafast electronic circuits [1], optical switches [2], and high-speed communication systems [3].

The carrier transport properties of a material are intrinsically linked to its atomic structure, which can range from amorphous to crystalline or polycrystalline. Tailoring the crystalline structure of semiconductors can significantly impact carrier recombination dynamics and mobility, thus influencing the efficiency of ultrafast devices. In particular, materials with short recombination times, on the order of a few picoseconds, are highly desirable for generating and manipulating fast electrical and optical signals [4].

Amongst the most widely used semiconductor materials for THz (opto-) electronics are the low-temperature-grown (LTG) GaAs and InGaAs, that have been widely used for devices operating in the near-infrared (800nm) and telecommunication bands (1550nm) respectively. Due to their direct bandgap and light effective electron mass, these III-V semiconductor group materials have exceptional electronic properties, including good carrier mobility and ultrafast recombination dynamics [5], [6]. In addition to these materials, Germanium (Ge) has been widely explored in recent years for THz device applications [7], [8], due to its low-cost and compatibility with traditional

semiconductor manufacturing processes. Due to its indirect bandgap, carrier dynamics in Ge are slow and therefore not suitable for THz electronics. Introducing, however, defects in the semiconductor’s crystalline structure via heavy ion bombardment can result in picosecond (ps) scale dynamics, thus making it suitable for high frequency electronics.

Indeed, these platforms have enabled early high frequency THz electronic and opto-electronic devices, however due to the very specialised techniques for preparing these materials, they are largely inaccessible for wider scientific and commercialisation purposes. This is because high quality LTG-GaAs, InGaAs or Ge require dedicated and expensive epitaxial techniques that are not widely available for large scale THz electronics. An alternative to these high-quality materials that was shown to be suitable for THz electronics, is the non-crystalline, amorphous-Ge. Opto-electronic devices made of this material have shown ultra-wide THz bandwidth up to 70THz [9]. Amorphous Ge, although being a lower-quality material, can be deposited on any substrate using cheap and widely available physical evaporation techniques that can be found in most nanofabrication facilities.

In this work, we present the first systematic study of Ge thin films grown by physical evaporation, and we investigate the influence of growth temperature on its carrier dynamics, mobility and generation of THz pulses. Depending on the substrate temperature, Ge can have amorphous, mono-crystalline or poly-crystalline lattice structure [10]. By varying the deposition temperature, we demonstrate that Ge can exhibit ultrafast carrier recombination even in its amorphous phase. We show that photoconductive switches (PCS) fabricated on Ge grown at different temperatures can generate electrical pulses in the ps range, highlighting the potential of Ge for ultrafast optoelectronic applications.

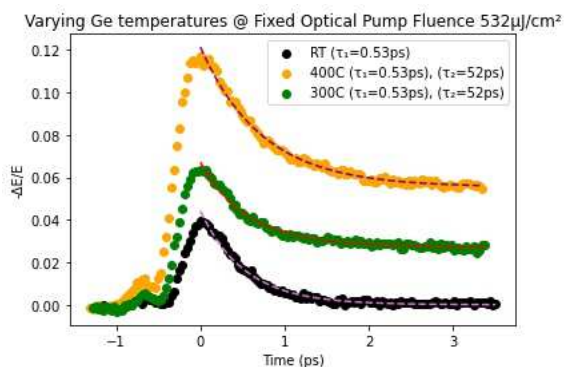


Fig. 1. OPTP results for RT, 300°C, and 400°C grown Ge samples indicating ps carrier recombination times. Simple and double exponential fits appear in dashed line.

II. RESULTS

Germanium films with 800nm thickness were fabricated using electron-beam physical evaporation, on top of semi-insulating GaAs substrates. To access and study the impact of all three of Germanium's crystal structures on the carrier properties, we have performed several depositions at various substrate temperatures, including room temperature and ranging from 200°C to 400°C in 50°C increments.

To measure the carrier dynamics, we performed optical pump THz-probe (OTPT) experiments at a pump wavelength of 800nm, for which the Ge absorption length is 64nm. As shown in Fig. 1, our measurements indicate that carriers in Ge have a short lifetime, comparable to that of LTG-GaAs. For the Ge films deposited at room temperature (RT), the dynamics show a single exponential behaviour with recombination times of approximately 0.53ps. At higher growth temperatures, the carrier dynamics still show fast dynamics; however, we observe a bi-exponential behaviour with characteristic times of 0.53ps and 52ps respectively. This difference in recombination times is attributed to variations in the crystalline structure of Ge. At lower temperatures, the material has no long-range crystalline order and has several defect states, or "traps", that can contribute towards fast carrier recombination times. In contrast, at higher temperatures, the material begins to crystallise and thus reducing the density of traps and leading to longer recombination times, as observed in the 300°C and 400°C samples (Fig. 1).

To examine the suitability of evaporated Ge for THz electronics and opto-electronics, we have fabricated planar photo-conductive switches (PCS) on Ge films produced at various temperatures, as shown in Fig. 2 (inset). The 780nm pump and probe beams were focused on the PCSs with 15 mW average power while one PCS was biased at 10V. THz signal detection was facilitated through the second PCS via standard lock-in amplifier techniques. Due to the high intrinsic carrier concentration of Ge, which is a result of the relatively low bandgap or shallow traps, it is worth noting that, for our geometry, the measured dark resistance is in the order of tens of k Ω , which is much lower than that of LT GaAs.

As illustrated in Fig. 2, the generated THz signals are in the picosecond range ($\sim$ 2ps). The ultrashort pulse generation result is consistent even for the devices made on Ge evaporated at

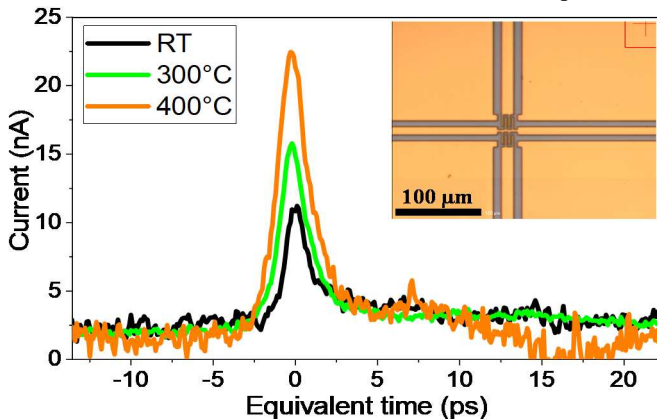


Fig. 2. Pump-probe results for RT, 300°C and 400°C grown Ge samples. Inset: Microscope picture of PCS.

higher temperatures, for which the carrier dynamics show a long recombination tail (see Fig. 1). In addition, we observe that the amplitude of the generated signal is higher for the devices evaporated at moderate temperatures, as shown in the green and orange curves in Fig. 2. We attribute this higher signal to an increased carrier mobility in Ge films evaporated at higher temperatures. This is because, as the substrate evaporation temperature increases, Ge transitions from an amorphous state to a more crystalline state with lower trap density and charge carriers can therefore travel a longer distance before they recombine.

III. SUMMARY

In conclusion, we have presented an alternative approach to the development of THz electronic and opto-electronic devices, which is cost-effective, and can be accessible through standard nanofabrication facilities. We showed that thin Ge layers evaporated at moderate temperatures, exhibit sub-picosecond carrier dynamics and can be used to generate THz pulses with picosecond duration. Our results show that, even though at higher deposition temperatures Ge has longer carrier dynamics, it does not significantly affect the duration of the THz pulses. On the contrary, we observe that the amplitude is increased due to higher carrier mobility. We anticipate that evaporated Ge can open up a wide range of possibilities for THz technology and enable the integration of high frequency electronics in systems where traditional epitaxial methods is not an option.

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